

**The Economic Impact of Cyberattacks on Businesses:
Data-Breach Costs and Cybersecurity Investment Decisions**

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Abstract

Cybersecurity is an important economic and resource allocation problem because cyberattacks incur direct costs of responding to them, as well as indirect business losses. This paper reviews the published global average cost of data breaches from 2021 to 2024 and discusses the potential implications of these financial impacts on decisions regarding investment in security. A clear descriptive analysis of four IBM annual report estimates shows an increase from \$4.24 million in 2021 to \$4.88 million in 2024, a cumulative increase of around 15.1%. A benchmark check using 2023 instead of 2021 still shows a 9.7% rise through 2024. Hypothetical expected-loss scenarios show the impact of the assumed probability of incidents on the estimated financial exposure. The analysis is more descriptive than causal as the underlying estimates are not panel data at the firm level and the probabilities are assumptions. Managers should use external cost benchmarks with incident probabilities, controls and business-interruption estimates specific to their organization.

1. Business Question

The primary research question is: What is the effect of cyberattacks on the financial performance of firms and to what extent do potential losses affect decisions on cybersecurity investments? This question is important for chief financial officers, chief information security officers, risk managers, insurers and investors. A cyber incident can result in immediate

investigation, restoration, legal assistance, notification and regulatory response spending. Disruption can also slow sales and production while payroll and fixed costs continue. Longer term effects could include customer turnover, higher insurance costs and more security investment. The consequences will differ for different industries and different companies, so the average reported is a benchmark, not a forecast for every firm.

The decision to implement a cybersecurity control is a trade-off between the expected reduction in losses from that control and the total costs of implementing and running it. When credible threats have the potential to disrupt critical operations, it is rarely an efficient strategy to allocate no resources to cybersecurity. However, organizations may also waste limited resources on unnecessary or ineffective controls without an accurate assessment of risk exposure. This study does not aim to estimate all the economic consequences of ransomware, phishing, and intrusions. The study instead uses published average data-breach costs as a measurable indicator and ties that indicator to an illustrative investment decision. This scope retains the original broader business question but acknowledges that data breaches do not represent all types of cyberattacks.

2. Data

The core data set includes four published global average cost-of-a-data-breach estimates from IBM Security/Ponemon Institute for report years 2021-2024. The main measure in the analysis is the published global average cost of a data breach in nominal U.S. dollars (millions). The time variable is year. The published breach-cost figures are used to derive the annual percentage changes and the cumulative percentage change. These are survey benchmarks done annually, not the same businesses observed over time. These should not be taken as inflation-adjusted numbers, or as a random sample of all cyber incidents. The methodology and

organizations sampled for the report may vary across editions (IBM Security, 2021, 2022, 2023, 2024).

The paper also uses hypothetical annual incident probabilities of 5%, 10% and 20% solely to demonstrate an expected-loss framework. These probabilities are not from the IBM reports, and they are not forecasts of the risk of any firm. The paper also does not claim to measure prevention effectiveness, stock-market losses, ransomware payments, or the full social cost of cybercrime. The CSV data set and Python analysis script to replicate the calculations in this paper are available in the GitHub repository associated with this paper.

3. Method

The analysis is a descriptive comparison of the costs of breaches over time. For each year following 2021: $\text{Annual percentage change} = 100 \times (\text{reported average in current year} / \text{reported average in previous year} - 1)$. Cumulative change is based on the same expression from 2021 to 2024. This approach is appropriate for the small number of published summary figures as it communicates the direction and magnitude of changes in breach costs without over-claiming what the data can support. Since there are only four annual aggregate observations, regression analysis would not be a reliable means to provide statistical evidence. The analysis thus makes no claims about statistical significance or causation.

The paper uses an expected-loss calculation to show how the financial impact of a cyberattack can be used to inform cybersecurity investment decisions. The estimated annual loss is the product of the assumed annual probability of a cyberattack and the estimated financial loss of the incident. $\text{Expected Annual Loss} = \text{Assumed Annual Probability of a Cyberattack} \times \text{Estimated Financial Loss}$. The estimated financial impact of an incident is \$4.88 million, the published average breach cost for 2024, and three scenarios have been calculated based on this.

Although the estimated expected losses show possible financial risk, they should not be used to determine how much money a company should spend on cybersecurity measures. It is determined by the fraction of losses that a proposed control prevents, its implementation and operating costs, any risk transfer, and risk tolerance. More generally, expected annual benefit = baseline expected loss – residual expected loss after control implementation. A cybersecurity investment can be considered economically justified when its expected annual benefit exceeds its annualized cost, although other risk-management considerations may also affect the decision (Gordon & Loeb, 2002).

4. Results

Table 1 summarizes the published annual cost benchmarks and the percentage changes computed by the accompanying script. The series climbs each year, with the published increase from 2023 to 2024 being particularly large. In nominal terms, the difference between the reported 2021 and 2024 averages is \$0.64 million per breach. This is comparing report averages, not confirming that a specific company’s expected loss increased by the same amount.

Table 1. Global Average Cost of a Data Breach, 2021-2024

Year	Average Cost of a Data Breach	Change from Prior Year
2021	\$4.24 million	n/a
2022	\$4.35 million	2.6%
2023	\$4.45 million	2.3%
2024	\$4.88 million	9.7%

Source: IBM Cost of a Data Breach Reports (2021–2024). Percentage changes were calculated by the author using the reported annual global average breach costs.

The nominal published average increased by about 15.09% over the whole 2021 to 2024 comparison. The upward direction also suggests that financial exposure arising from breaches is worthy of consideration in discussions of capital allocation. However, movement may be attributed to inflation, changes in survey composition, incident characteristics, or reporting methodology. The data alone cannot tell which factor explains the rise or demonstrate that cyberattacks caused changes in company profitability.

Table 2. Scenario Analysis of Expected Annual Cyberattack Loss

Assumed Annual Probability of a Breach	Average Breach Cost	Estimated Expected Annual Loss
5%	\$4.88 million	\$244,000
10%	\$4.88 million	\$488,000
20%	\$4.88 million	\$976,000

Note: The 5%, 10%, and 20% probabilities are hypothetical scenarios used for sensitivity analysis and are not the observed probability of breach. Expected annual loss is the assumed probability of a breach times IBM's reported \$4.88 million global average breach cost in 2024.

To examine the possible effect of breach risk on cybersecurity investment decisions, a scenario analysis was conducted using IBM's reported 2024 global average breach cost of \$4.88 million. Because IBM data reports average financial costs of a breach, not the annual probability of an individual company experiencing a breach, three hypothetical annual breach probabilities of 5%, 10%, and 20% were used. These scenarios are not estimates of the real likelihood of a breach but are used to show how changes in assumed cyber risk affect expected financial loss. Expected annual loss was calculated as the product of the assumed likelihood of a breach and the average cost of \$4.88 million. This calculation results in an expected annual loss of \$488,000 at a hypothetical 10% annual probability. The illustrative expected annual benefit would be

\$244,000, not counting other benefits or costs, if a proposed program were expected to cut that probability in half and not change incident severity. Real decisions need believable organization-specific probability and effectiveness estimates.

5. Validation and Limitation

A simple alternative-period validation is to repeat the direction-of-change calculation using 2023 as the baseline, instead of 2021. The published average rises from \$4.45 million in 2023 to \$4.88 million in 2024, or around 9.66%. The result is in the same direction as the full-window 15.09% increase, but the shorter comparison is of a different size. The Python script also performs a second numerical check by re-calculating each of the reported percentage changes directly from the CSV data, rather than using any manually entered calculations. These checks are to validate the descriptive calculations, not the representativeness of the source surveys.

The analysis has significant limitations. Four yearly summary observations do not suffice to support robust forecasting, formal statistical inference, or causal attribution. The numbers are averages of reported breach costs, not the distribution of losses. An average can be a poor characterization of rare extreme incidents and smaller incidents. Comparisons in nominal dollars also do not distinguish between general inflation and real change in the cost of breaches. There is no empirical return on cybersecurity investment that can be derived from hypothetical probability scenarios. A more robust follow-up study would collect company-level breach dates, disclose costs, revenue, operating income, industry, and firm size and compare firms before and after the incident to matched unaffected firms. Such a study could test the effects of disruption and profitability separately.

6. Recommendations

Firstly, managers should think of cybersecurity as enterprise risk management rather than as a stand-alone IT cost. Finance and security teams should keep a shared inventory of critical systems and estimate the cost of operational downtime, and model direct response costs along with potential reputational and legal exposures. The published global average provides an external discussion benchmark, but the firm's own loss distribution is the right basis for budgets.

Secondly, investment proposals must quantify the expected reduction in risk rather than using a large industry average. For every suggested control, including multi-factor authentication, backup recovery, detection and response, or employee training, decisionmakers need to assess both implementation and ongoing expenses. They should also think about relevant incident scenarios and estimate the extent to which the control could reduce the probability or financial impact of an incident. Sensitivity analysis should indicate if the decision is still attractive under low, medium and high-risk assumptions. Because security controls interact, benefits should not be added up without considering the overlapping risk reductions.

Thirdly, managers need to monitor outcomes after deployment. Useful internal metrics include recovery time, hours of business interruption, success of tested backup restoration and cost of actual incidents. The investment model gains more credibility by annual updating of assumptions than by a fixed forecast based on a survey average. Investors can also assess whether a company's cybersecurity governance and incident response capabilities are in line with its risk profile. However, the analysis does not assume that increased cybersecurity spending will lead to higher profits. Overall, the descriptive evidence points to the serious nature of cyber losses. However, the optimal spending decision depends on the true exposure of the organization and the proven effectiveness of the controls.

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Appendix A

Supporting Data and Calculations

A1. Calculations Supporting Table 1

2021-2022: $(4.35 / 4.24 - 1) \times 100 = 2.6\%$

2022-2023: $(4.45 / 4.35 - 1) \times 100 = 2.3\%$

2023-2024: $(4.88 / 4.45 - 1) \times 100 = 9.7\%$

2021-2024 cumulative change: $(4.88 / 4.24 - 1) \times 100 = 15.09\%$

A2. Calculations Supporting Table 2

Assumed Annual Breach Probability	2024 Average Breach Cost	Calculation	Expected Annual Loss
5%	\$4.88 million	$0.05 * \$4.88\text{M}$	\$244,000
10%	\$4.88 million	$0.10 * \$4.88\text{M}$	\$488,000
20%	\$4.88 million	$0.20 * \$4.88\text{M}$	\$976,000

A3. Replication Materials

The CSV data and Python script to reproduce the calculations can be found in the GitHub repository in the Submission Links section.

Submission Links

GitHub repository: <https://github.com/sahirkhn4721/economic-impact-cyberattacks>